5.e) Genetic load and its relevance

Genetic load refers to the relative disadvantage a given population suffers from carrying certain disadvantageous genes.

It refers to difference from optimal or best case of fitness of given population.

$$\boxed{\text{Genetic load, } G = 1 - \overline{w}}$$

where $1 \rightarrow$ optimal fitness
$\overline{w} \rightarrow$ average fitness of population

Relevance

Superficially genetic load might sound alarming as it reduces the fitness of given species due to carrying such 'burden' of genes not suitable for current adaptation.

However, many anthropologists &

geneticists have identified its actual relevance for survival of species.

Genetic load by acting as filter help regulate the population of any given species in order to not breach the carrying capacity of environment.

Also, genetic load provides a species with such genes which might not be advantageous for adaptation but whose mutation or combination with other genes might help in better reproductive success in the future.

Genetic load is therefore important feature that needs to be studied across all human populations to estimate where they stand.